Final Spring Boot Interview Question List

1. Spring Core

1. What is Spring?
2. Why do we use Spring?
3. Why is Spring called a lightweight framework?
4. What are the major features of Spring?
5. What are the advantages of Spring?
6. What is IoC (Inversion of Control)?
7. What is Dependency Injection?
8. IoC vs Dependency Injection?
9. What is loose coupling?
10. What is tight coupling?
11. How does Spring achieve loose coupling?
12. What is a Spring Container?
13. What is a Spring Bean?
14. How is a Bean created?
15. What is ApplicationContext?
16. ApplicationContext vs BeanFactory?
17. What is component scanning?
18. What is @ComponentScan?
19. What is autowiring?
20. What are the types of Dependency Injection?
21. Constructor injection vs setter injection?
22. Why is constructor injection preferred?
23. What is @Autowired?
24. What is @Qualifier?
25. What is @Primary?
26. What happens when multiple beans of the same type exist?
27. What exception occurs when multiple beans match?
28. What is a stereotype annotation?
29. What is @Component?
30. What is @Service?
31. What is @Repository?
32. What is @Controller?
33. @Component vs @Service?
34. @Service vs @Repository?

35. @Component vs @Service vs @Repository?
36. What is @Configuration?
37. What is @Bean?
38. @Bean vs @Component?
39. What are Spring Bean scopes?
40. Singleton vs Prototype?
41. What is Bean lifecycle?
42. What is @PostConstruct?
43. What is @PreDestroy?
44. What is eager initialization?
45. What is lazy initialization?
46. What is @Lazy?
47. XML Configuration vs Java Configuration?
48. What is @Profile?
49. Why do we use Spring Profiles?
50. What is @Value?
51. What is @PropertySource?

2. Spring MVC

52. What is Spring MVC?
53. What is MVC architecture?
54. What is DispatcherServlet?
55. What is a Controller?
56. What is @Controller?
57. What is @RestController?
58. @Controller vs @RestController?
59. What is @RequestMapping?
60. @GetMapping vs @PostMapping?
61. What is @PutMapping?
62. What is @DeleteMapping?
63. What is @PathVariable?
64. What is @RequestParam?
65. What is @RequestBody?
66. What is @ResponseBody?
67. What is ResponseEntity?
68. How does a request flow through Spring MVC?

3. Spring Boot

69. What is Spring Boot?

70. Spring vs Spring Boot?
71. Why is Spring Boot preferred?
72. What are the advantages of Spring Boot?
73. What is Auto Configuration?
74. What is @SpringBootApplication?
75. Which annotations are present inside @SpringBootApplication?
76. What are Spring Boot Starter Dependencies?
77. What is a Starter Dependency?
78. What is an Embedded Server?
79. Which embedded server does Spring Boot use by default?
80. What is Embedded Tomcat?
81. What is application.properties?
82. What is application.yml?
83. application.properties vs application.yml?
84. How do you configure a database in Spring Boot?
85. How do you change the server port?
86. What is dependency management?
87. What are Spring Profiles?
88. What is @Profile?
89. What is externalized configuration?
90. What is Spring Boot Actuator?
91. What are Spring Boot Runners?
92. CommandLineRunner vs ApplicationRunner?
93. How does Spring Boot start an application?

4. Spring Data JPA

94. What is Spring Data JPA?
95. Why do we use Spring Data JPA?
96. Difference between JPA and Spring Data JPA?
97. What is a Repository?
98. What is CrudRepository?
99. What is PagingAndSortingRepository?
100. What is JpaRepository?
101. CrudRepository vs JpaRepository?
102. JpaRepository vs CrudRepository?
103. Why do we create Repository Interfaces?
104. How does Spring Data create Repository implementations?
105. Who implements JpaRepository methods?
106. What is save()?

107. What is findById()?
108. What is findAll()?
109. What is deleteById()?
110. What are Derived Query Methods?
111. How does findByName() work?
112. What is @Query?
113. What is JPQL?
114. JPQL vs Native SQL?
115. JPQL vs Native Query?
116. What is Pagination?
117. How is Sorting implemented?
118. What is Sorting?
119. How do you update using Spring Data JPA?
120. How do you delete using Spring Data JPA?

5. REST API

121. What is REST?
122. What is a REST API?
123. What are REST principles?
124. GET vs POST?
125. PUT vs PATCH?
126. What is DELETE?
127. What are HTTP Status Codes?
128. 200 vs 201?
129. 400 vs 401 vs 403 vs 404?
130. What is 500?
131. What is Idempotency?
132. What is JSON?
133. How does Spring Boot convert Java objects to JSON?
134. What is Jackson?
135. How do you handle exceptions globally?
136. What is @ControllerAdvice?
137. What is @ExceptionHandler?

6. Validation

138. What is Validation?
139. What is Bean Validation?
140. What is @Valid?
141. What is @NotNull?

142. @NotBlank vs @NotEmpty?
143. What is @Size?
144. How do you handle validation errors?
145. Why should validation be performed on API input?

7. Spring AOP

146. What is AOP?
147. Why do we use AOP?
148. What is a Cross-Cutting Concern?
149. What is an Aspect?
150. What is a Join Point?
151. What is Pointcut?
152. What is Advice?
153. What are the types of Advice?
154. What is @Before?
155. What is @After?
156. What is @Around?

8. Spring Testing

157. What is Unit Testing?
158. What is Integration Testing?
159. What is JUnit?
160. What is Mockito?
161. @Mock vs @MockBean?
162. What is @SpringBootTest?
163. What is @WebMvcTest?
164. Why do we use mocks?

9. Transactions

165. What is a Transaction?
166. What are ACID Properties?
167. What is @Transactional?
168. Why do we use @Transactional?
169. Commit vs Rollback?
170. What happens if an exception occurs inside a Transaction?